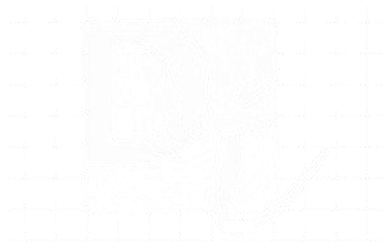


高级语言C++程序设计

Lecture 9 类和对象

简单数据结构

链表 (Linked List)



结点 (Node)

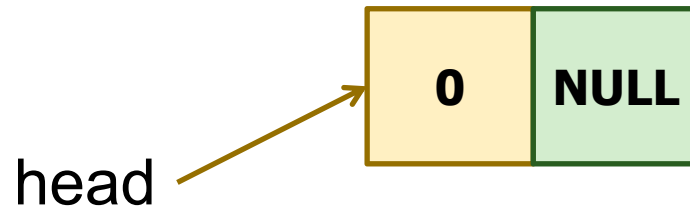
```
class Node{  
    public:  
        int data;  
        Node* next;  
        Node(int i) {  
            data = i;  
            next = NULL;  
        }  
};
```

Node



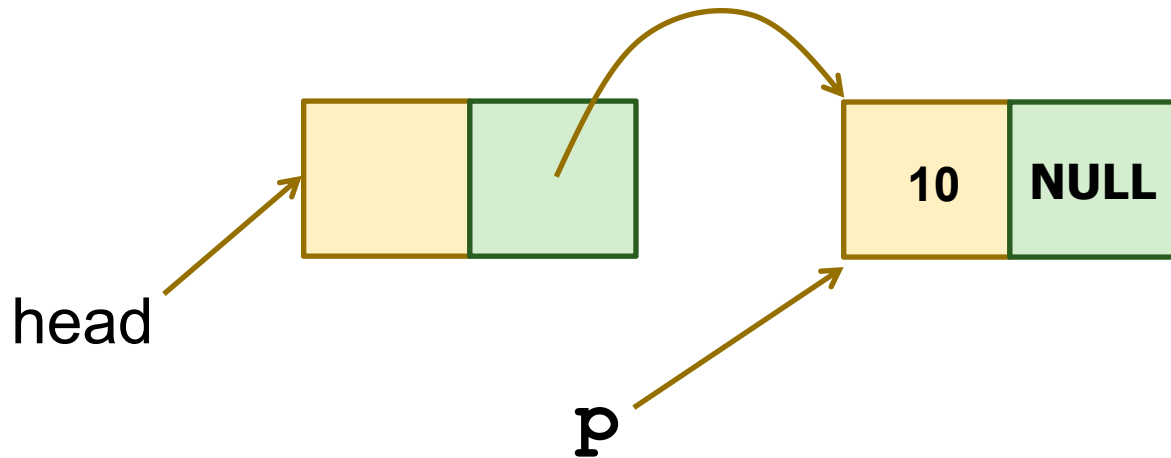
创建头结点

```
Node *head = new Node(0);
```



插入第一个结点

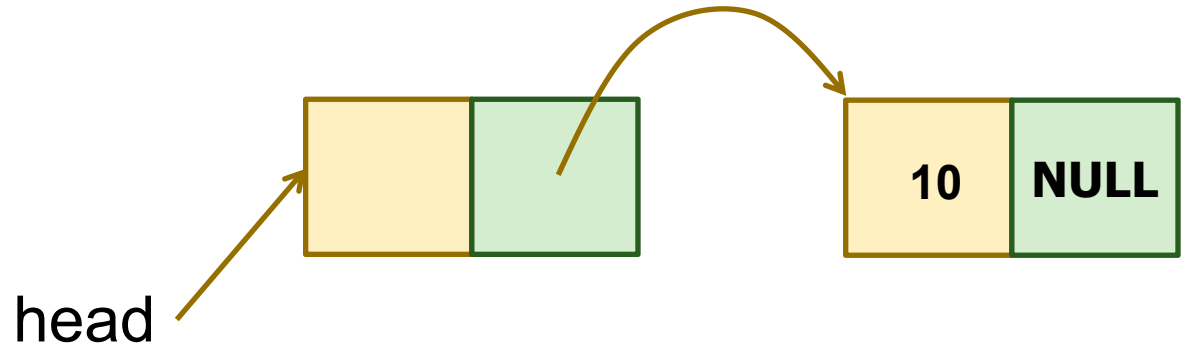
```
Node *p = new Node(10);  
p->next = head->next;  
head->next = p;
```



插入第一个结点

```
void Insert(int k, int data) {  
    Node *newNode = new Node(data);  
    Node *p = head;  
    for(int i=1; i<k; i++)  
        p = p->next;  
    newNode->next = p->next;  
    p->next = newNode;  
}
```

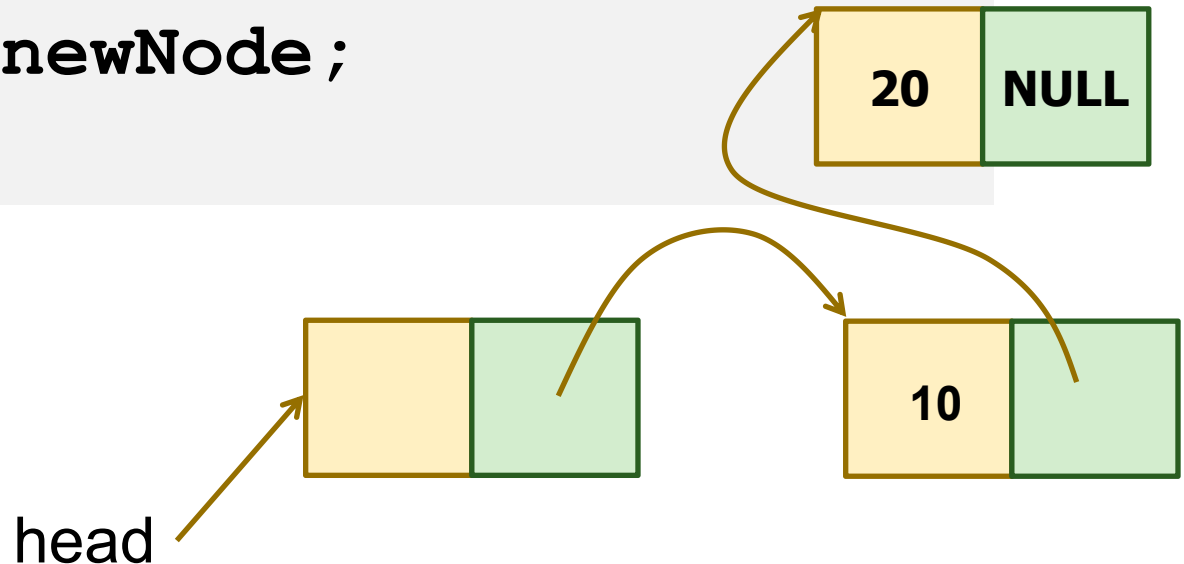
Insert(1, 10);



插入第二个结点(函数实现)

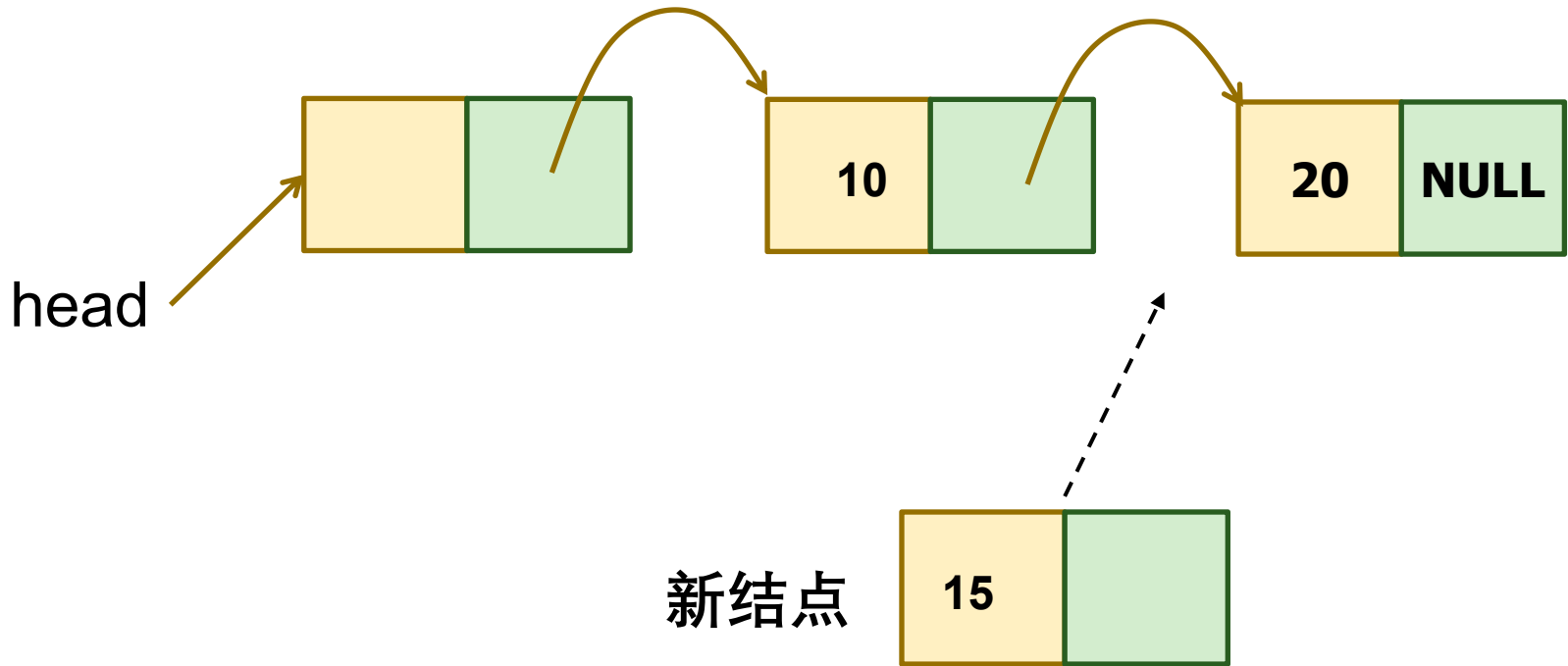
```
void Insert(int k, int data) {  
    Node *newNode = new Node(data);  
    Node *p = head;  
    for(int i=1; i<k; i++)  
        p = p->next;  
    newNode->next = p->next;  
    p->next = newNode;  
}
```

Insert(2, 20);



在第k位置插入结点

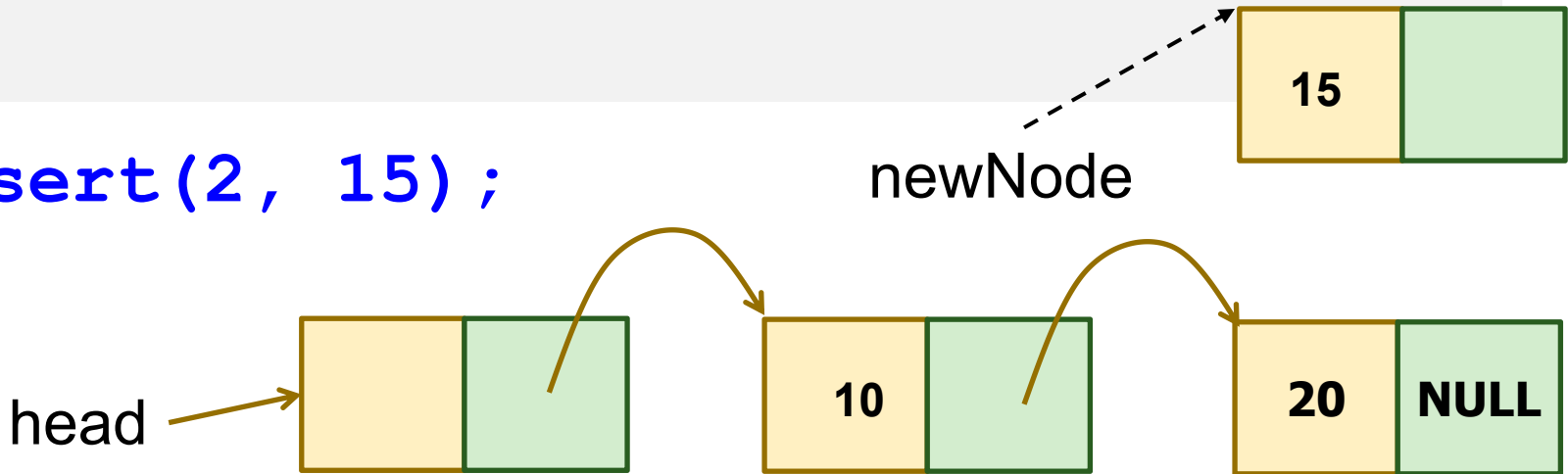
假设链表有2个node，在第2个node位置插入一个新的结点，新结点的元素为15



在第k位置插入结点

```
void Insert(int k, int data) {  
    Node *newNode = new Node(data);  
    Node *p = head;  
    for(int i=1; i<k; i++)  
        p = p->next;  
    newNode->next = p->next;  
    p->next = newNode;  
}
```

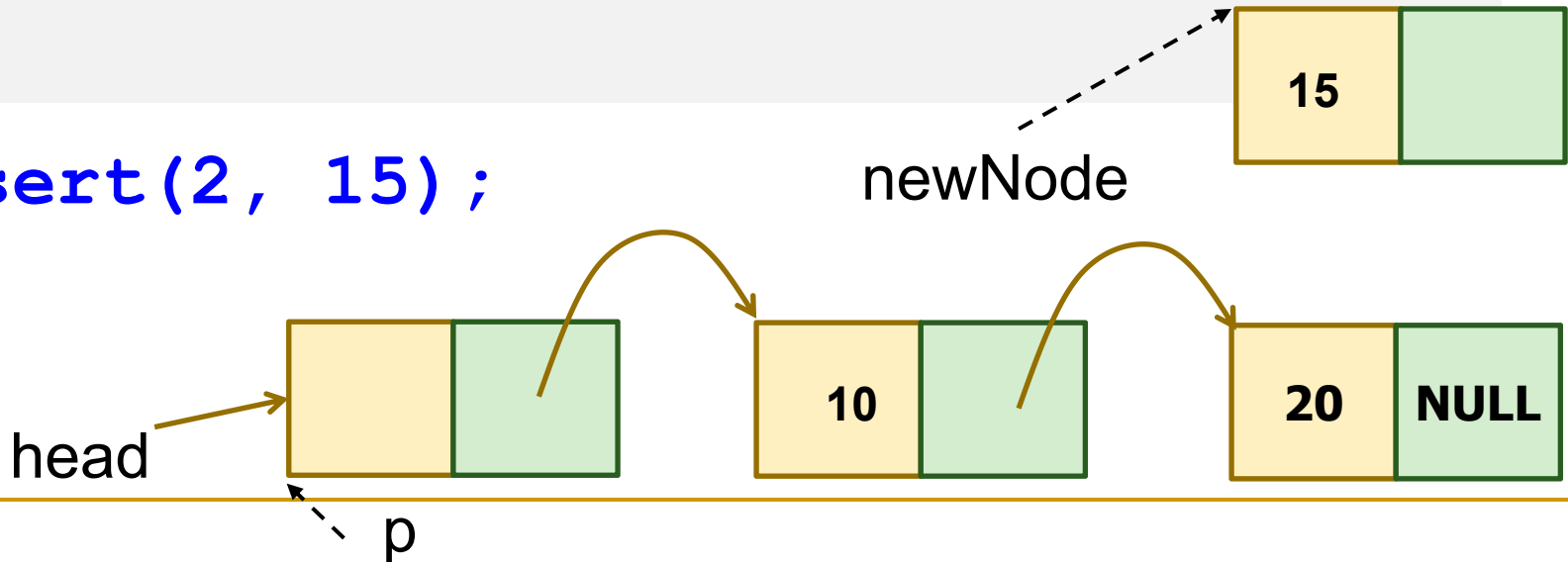
Insert(2, 15);



在第k位置插入结点

```
void Insert(int k, int data) {  
    Node *newNode = new Node(data);  
    Node *p = head;  
    for(int i=1; i<k; i++)  
        p = p->next;  
    newNode->next = p->next;  
    p->next = newNode;  
}
```

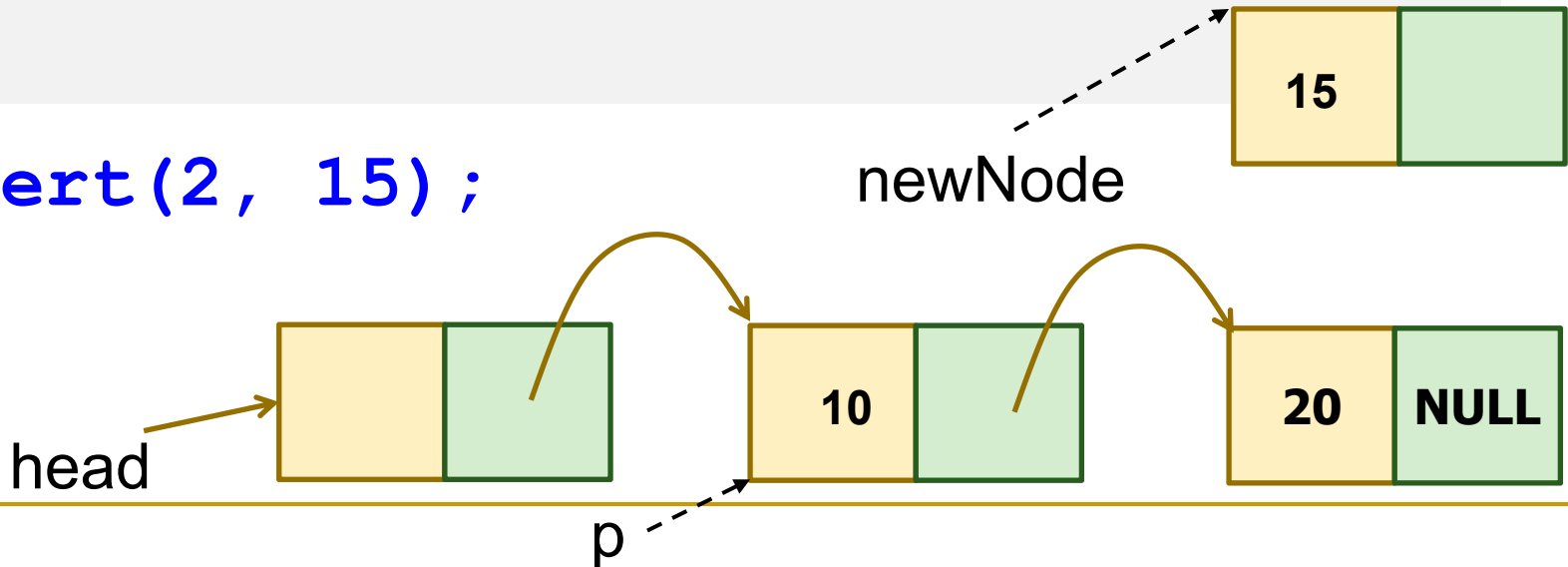
Insert(2, 15);



在第k位置插入结点

```
void Insert(int k, int data) {  
    Node *newNode = new Node(data);  
    Node *p = head;  
    for(int i=1; i<k; i++)  
        p = p->next;  
    newNode->next = p->next;  
    p->next = newNode;  
}
```

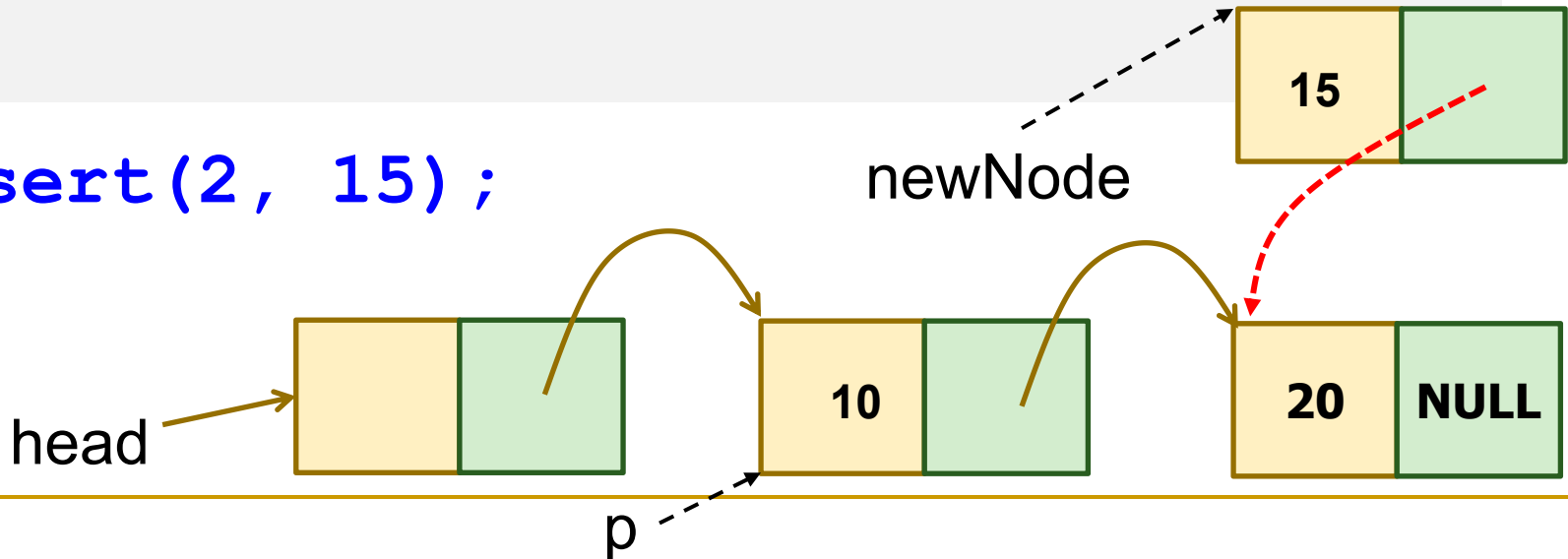
Insert(2, 15);



在第k位置插入结点

```
void Insert(int k, int data) {  
    Node *newNode = new Node(data);  
    Node *p = head;  
    for(int i=1; i<k; i++)  
        p = p->next;  
    newNode->next = p->next;  
    p->next = newNode;  
}
```

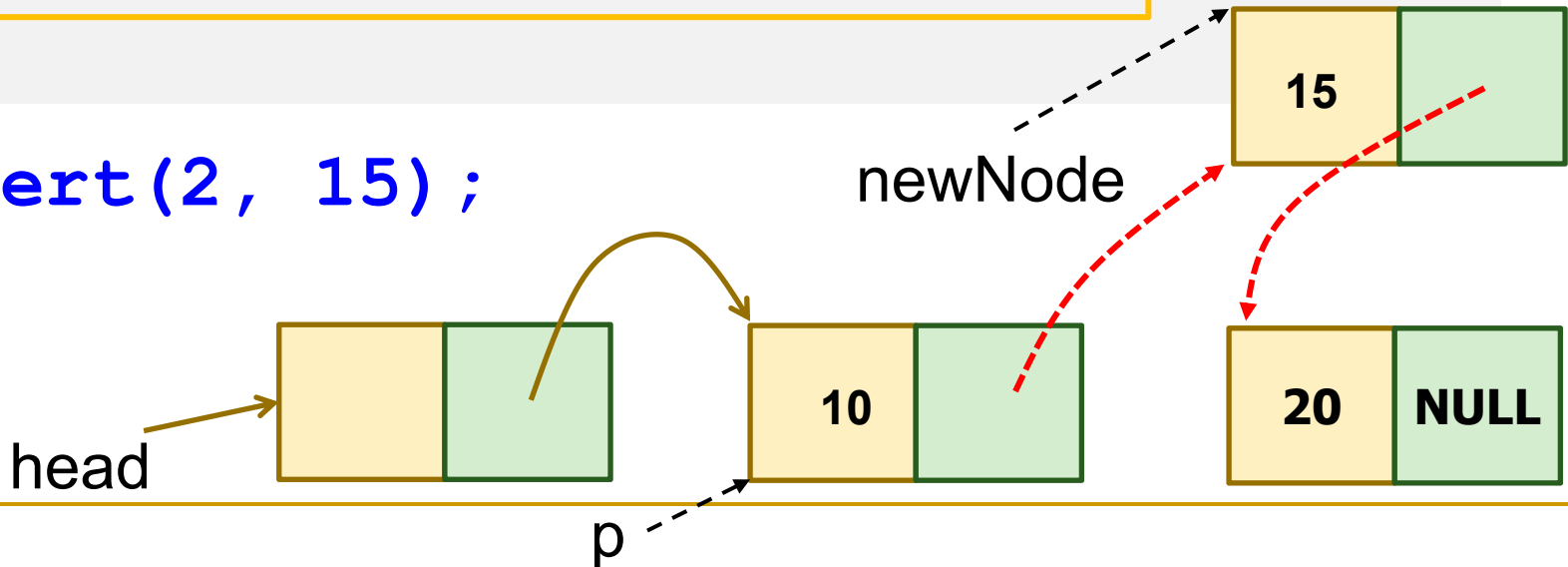
Insert(2, 15);



在第k位置插入结点

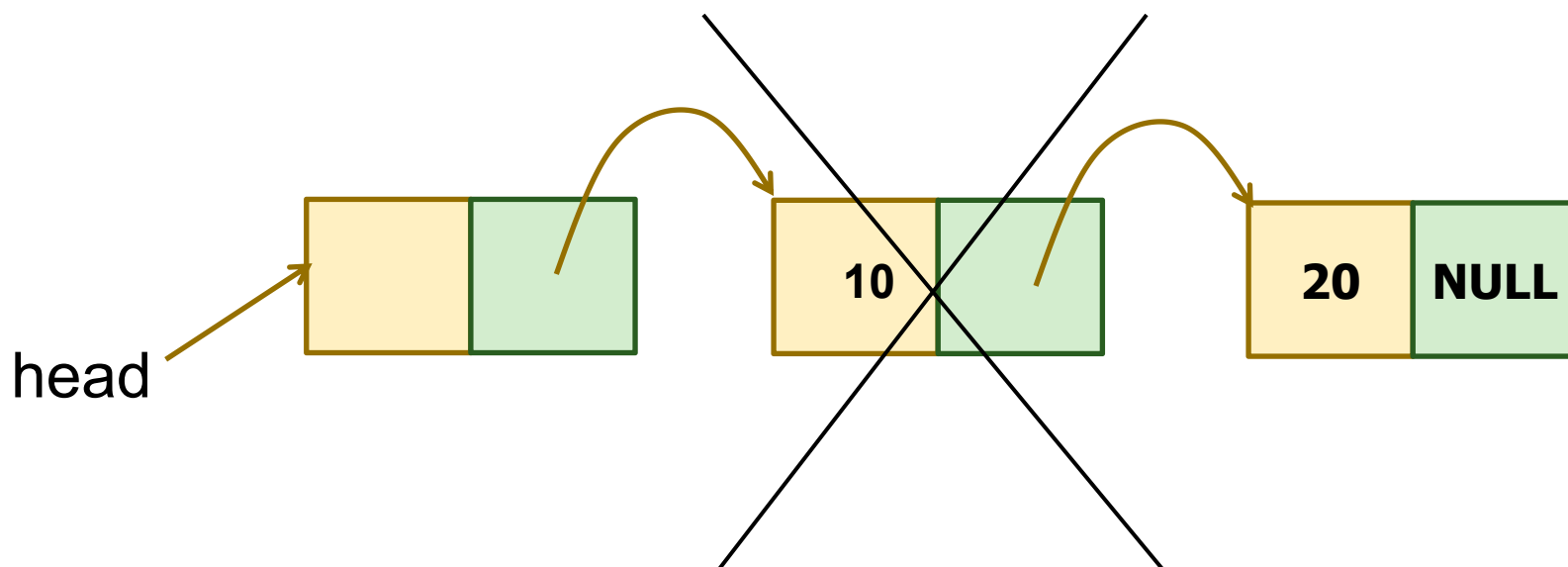
```
void Insert(int k, int data) {  
    Node *newNode = new Node(data);  
    Node *p = head;  
    for(int i=1; i<k; i++)  
        p = p->next;  
    newNode->next = p->next;  
    p->next = newNode;  
}
```

Insert(2, 15);



删除第k个结点

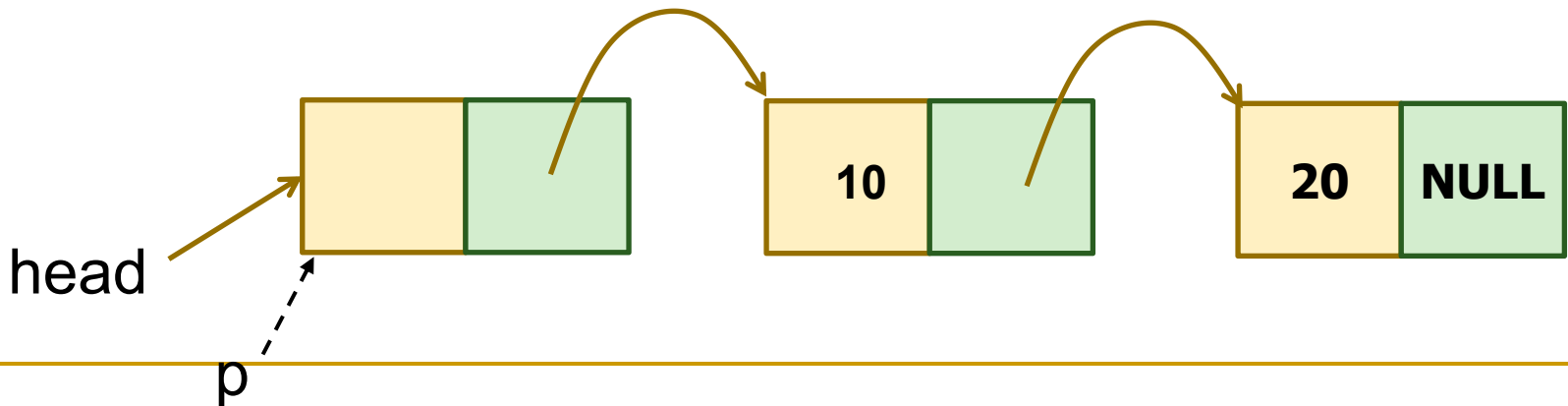
假设链表有2个node，删除第1个node



删除第k个结点

```
void Delete(int k) {  
    Node *p = head;  
    for (int i=1;i<k;i++)  
        p = p->next;  
    Node *q = p->next;  
    p->next = q->next;  
    delete q;  
}
```

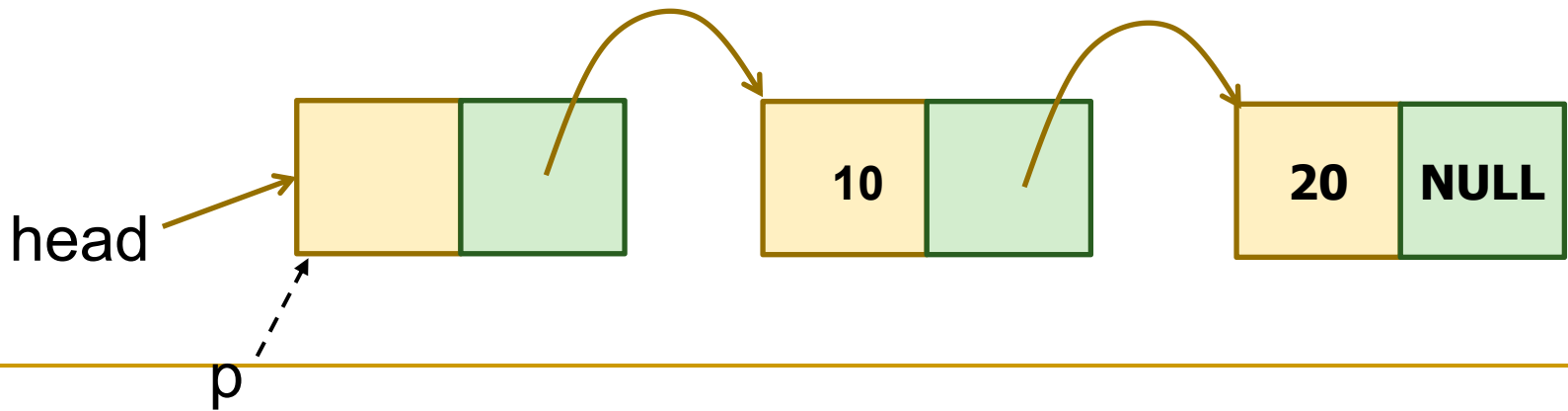
Delete(1);



删除第k个结点

```
void Delete(int k) {  
    Node *p = head;  
    for (int i=1;i<k;i++)  
        p = p->next;  
    Node *q = p->next;  
    p->next = q->next;  
    delete q;  
}
```

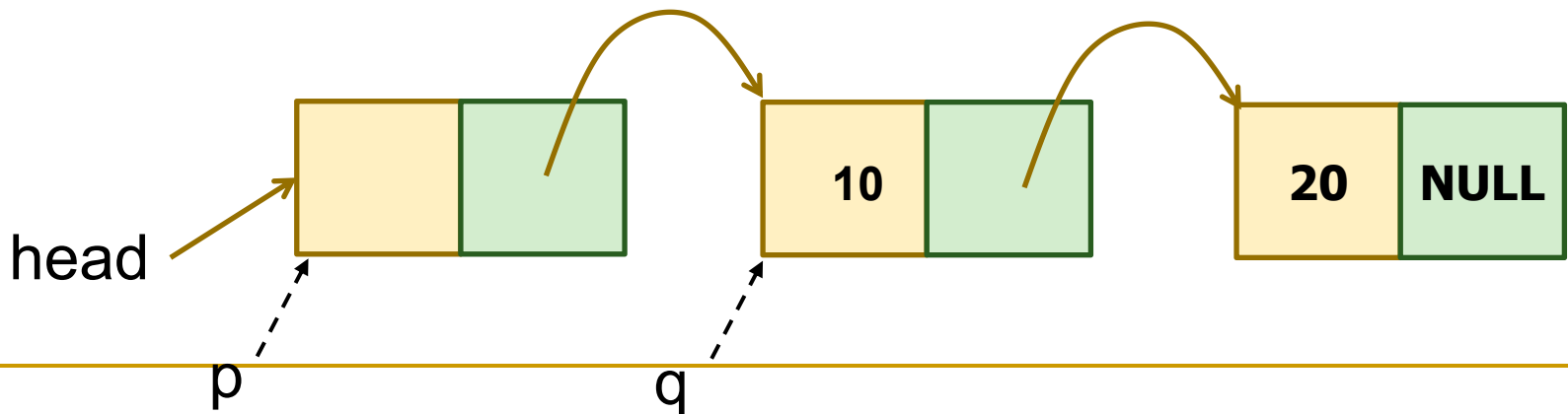
Delete(1);



删除第k个结点

```
void Delete(int k) {  
    Node *p = head;  
    for (int i=1;i<k;i++)  
        p = p->next;  
    Node *q = p->next;  
    p->next = q->next;  
    delete q;  
}
```

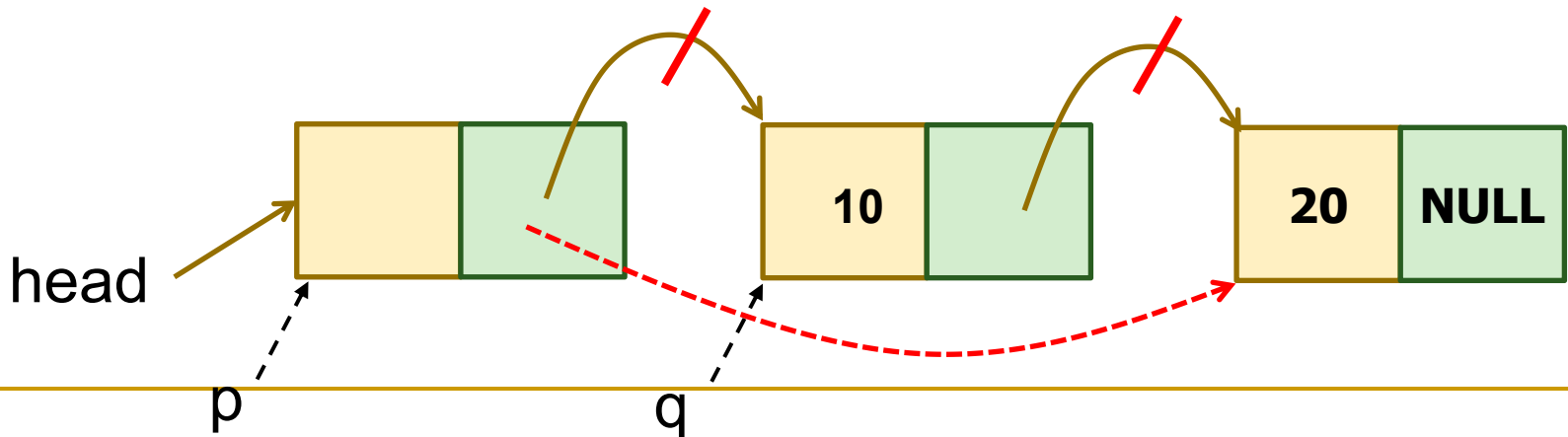
Delete(1);



删除第k个结点

```
void Delete(int k) {  
    Node *p = head;  
    for (int i=1;i<k;i++)  
        p = p->next;  
    Node *q = p->next;  
    p->next = q->next;  
    delete q;  
}
```

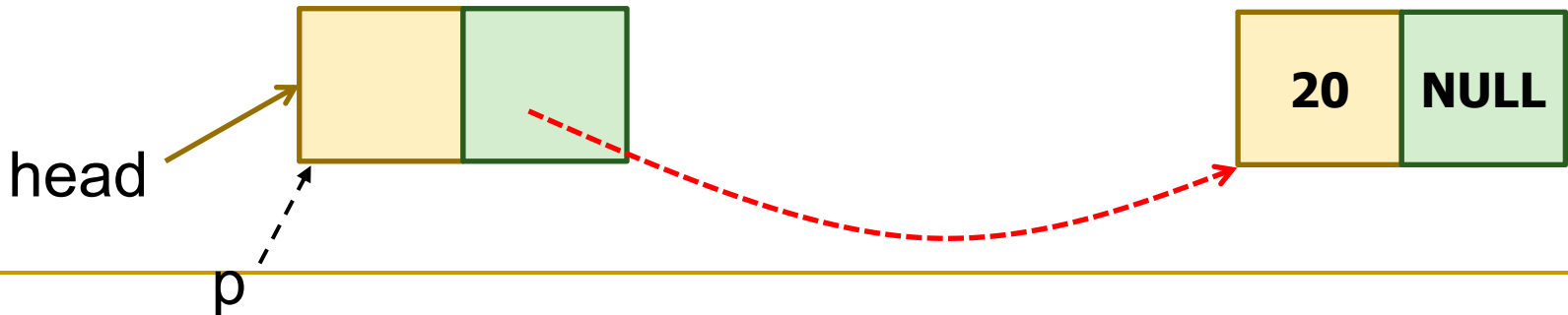
Delete(1);



删除第k个结点

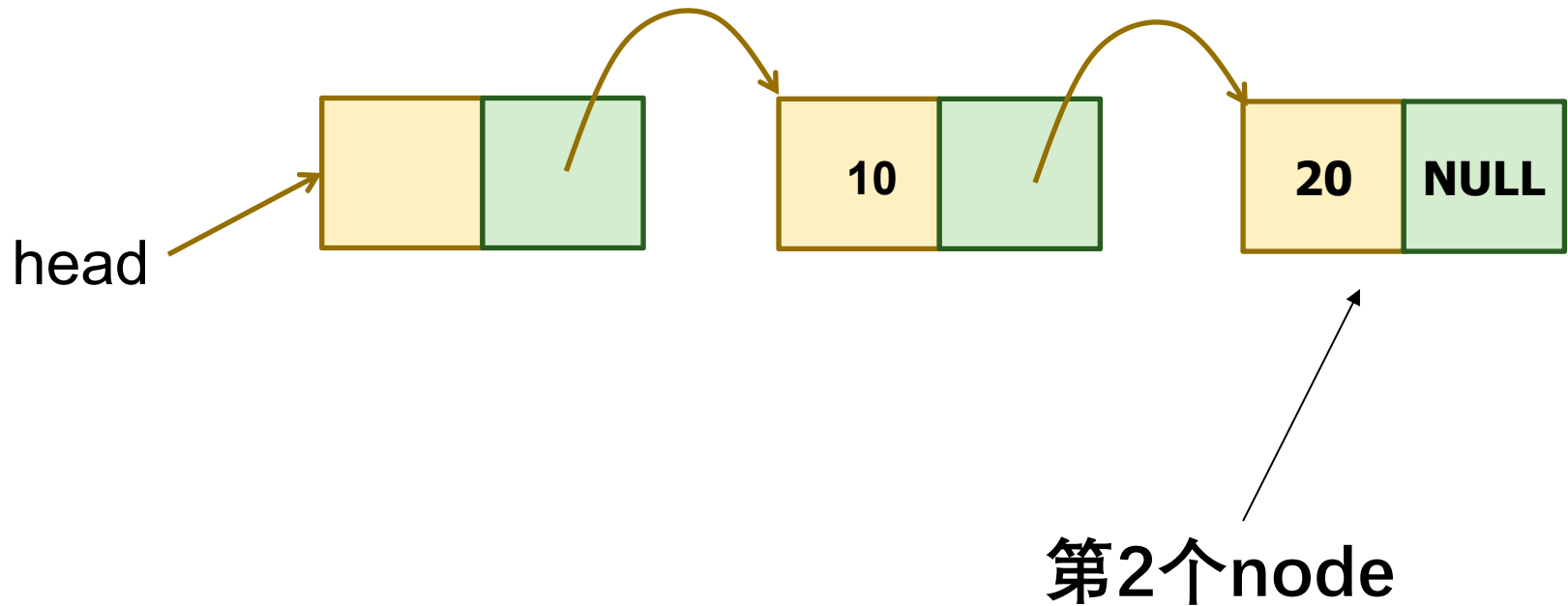
```
void Delete(int k) {  
    Node *p = head;  
    for (int i=1;i<k;i++)  
        p = p->next;  
    Node *q = p->next;  
    p->next = q->next;  
    delete q;  
}
```

Delete(1);



查找第k个结点

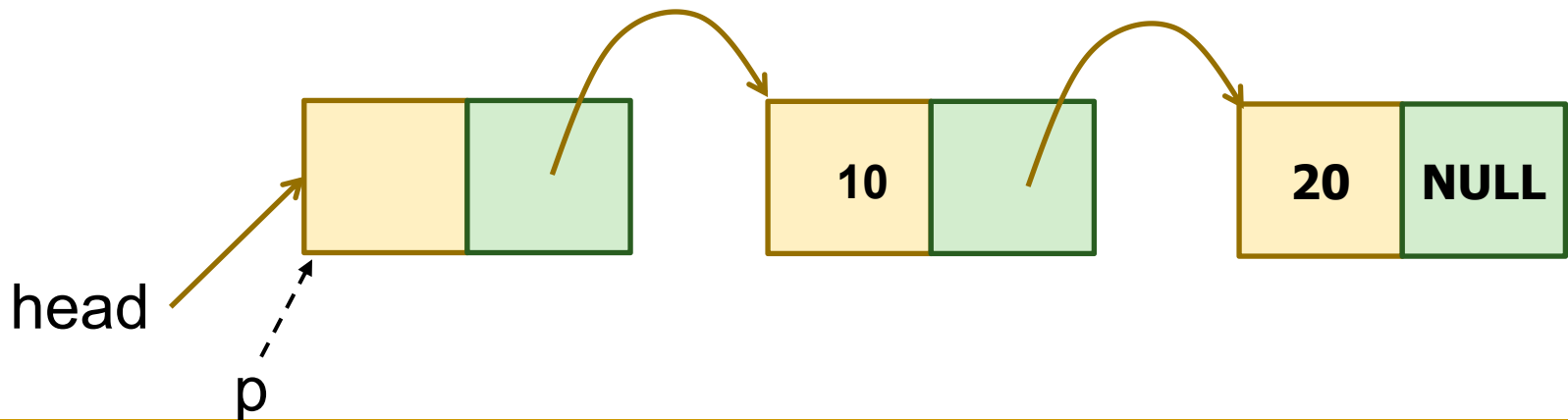
假设链表有2个node，查找第2个node，返回值



查找第k个结点

```
int Find(int k) {  
    Node *p = head;  
    for (int i=1;i<k;i++)  
        p = p->next;  
    Node *q = p->next;  
    return q->data;  
}
```

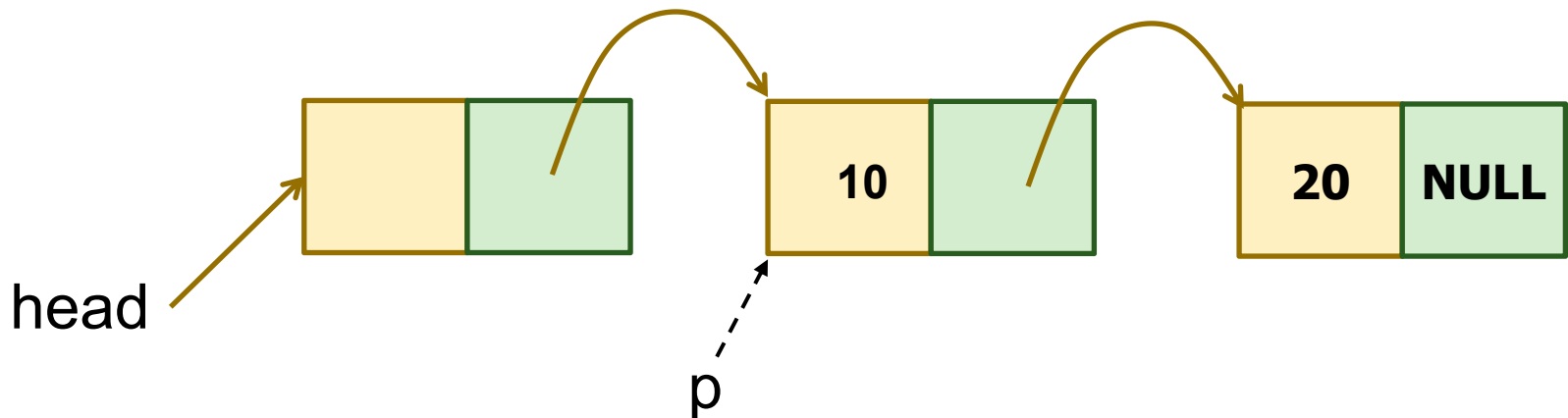
Find(2) ;



查找第k个结点

```
int Find(int k) {  
    Node *p = head;  
    for (int i=1;i<k;i++)  
        p = p->next;  
    Node *q = p->next;  
    return q->data;  
}
```

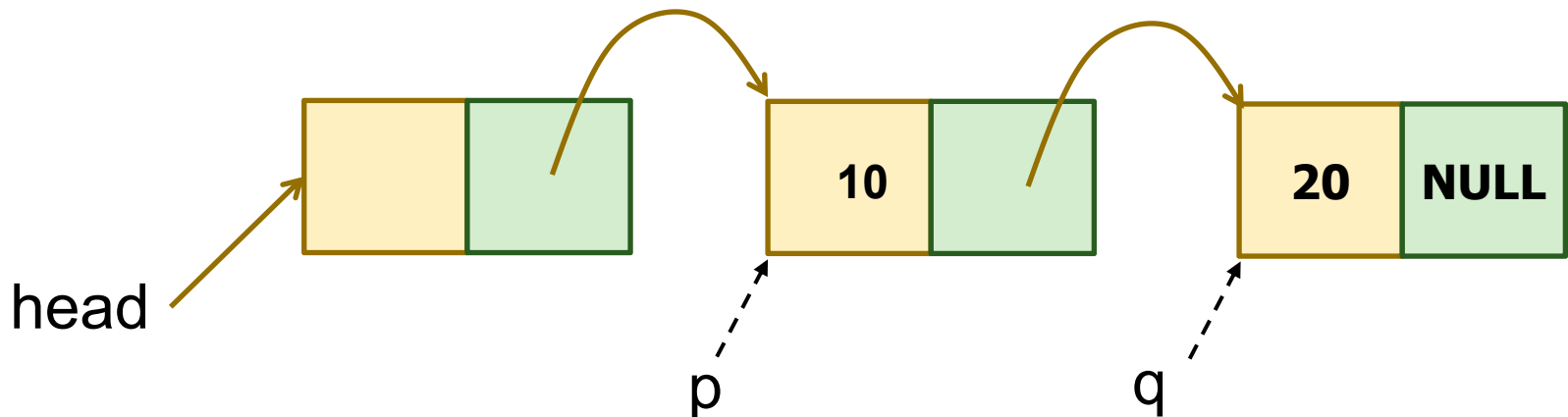
Find(2) ;



查找第k个结点

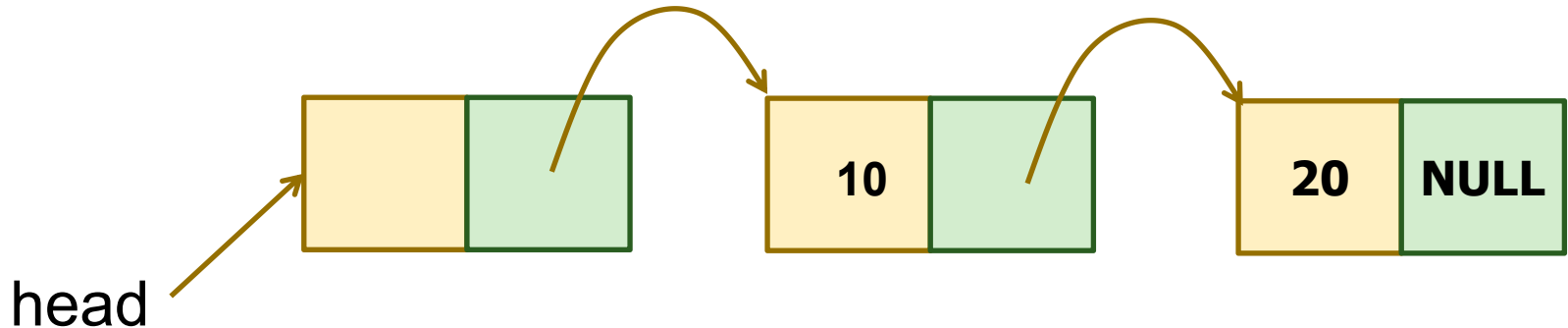
```
int Find(int k) {  
    Node *p = head;  
    for (int i=1; i<k; i++)  
        p = p->next;  
    Node *q = p->next;  
    return q->data;  
}
```

Find(2);

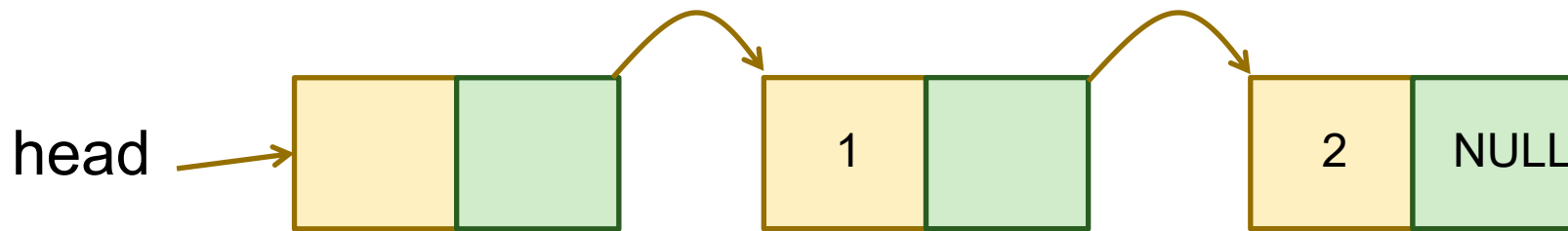


遍历所有结点

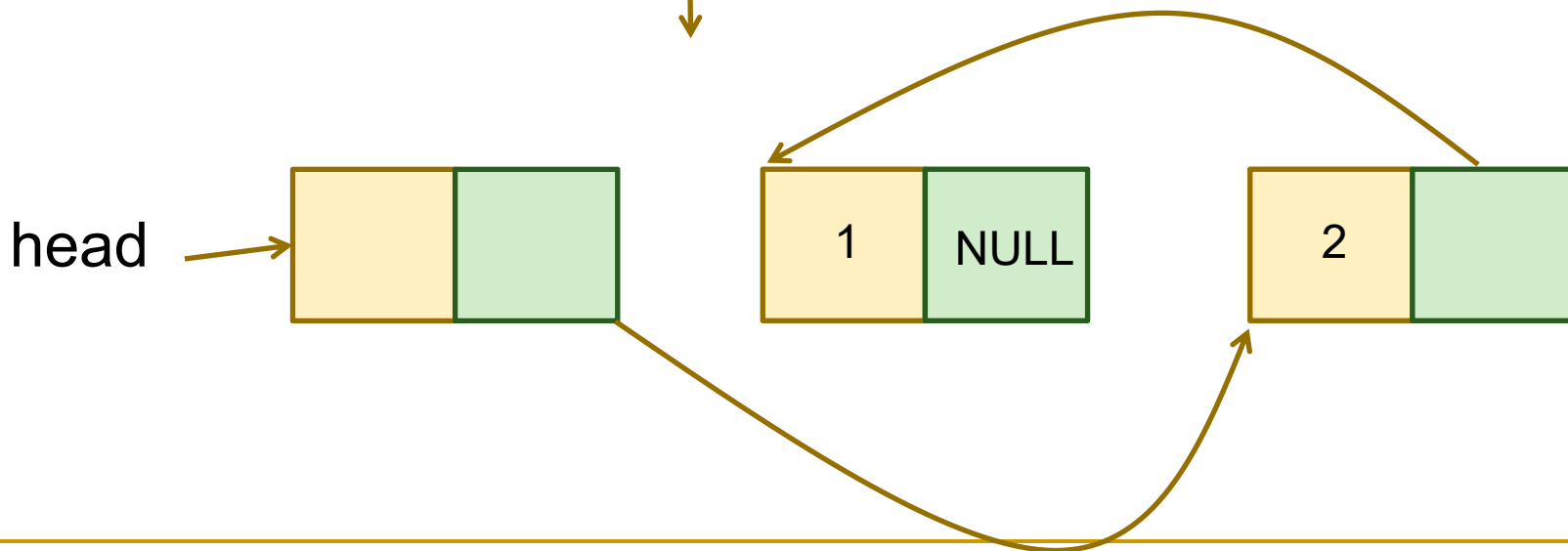
```
void Print() {  
    Node *p = head;  
    while(p) {  
        p = p->next;  
        if (p != NULL)  
            cout<<p->data<<endl;  
    }  
}
```



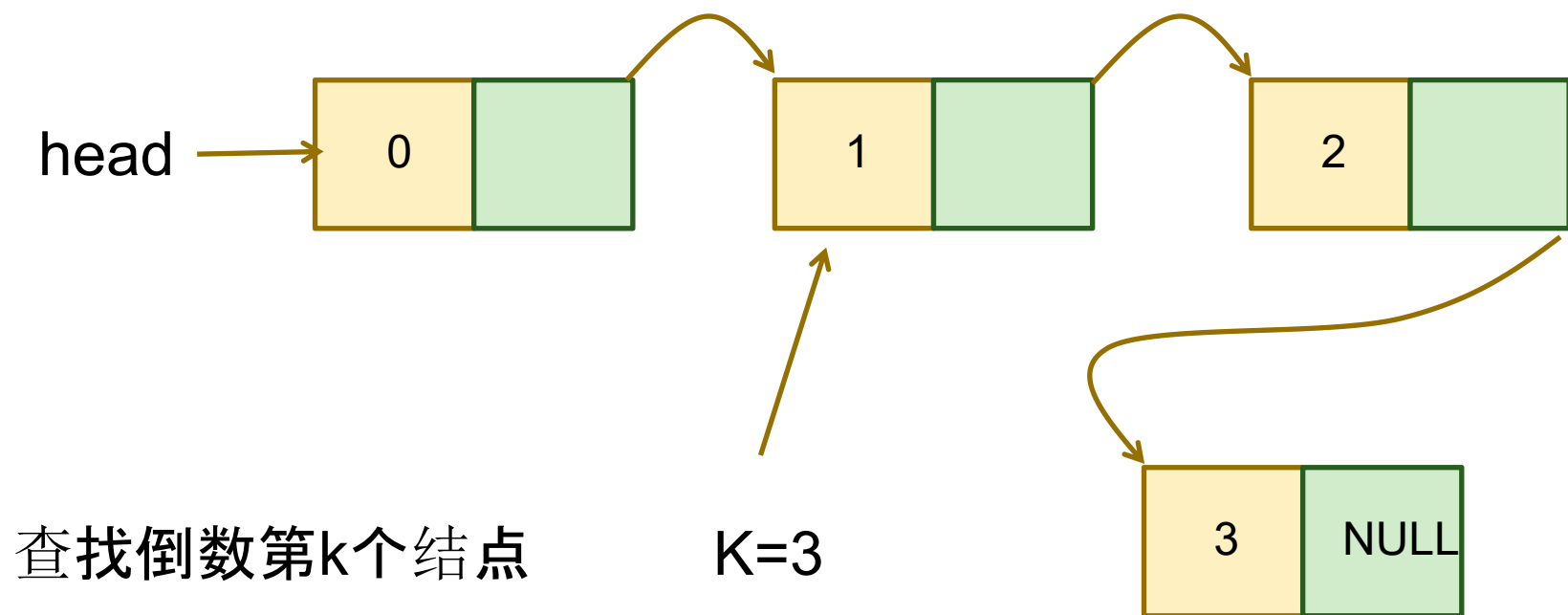
链表反转



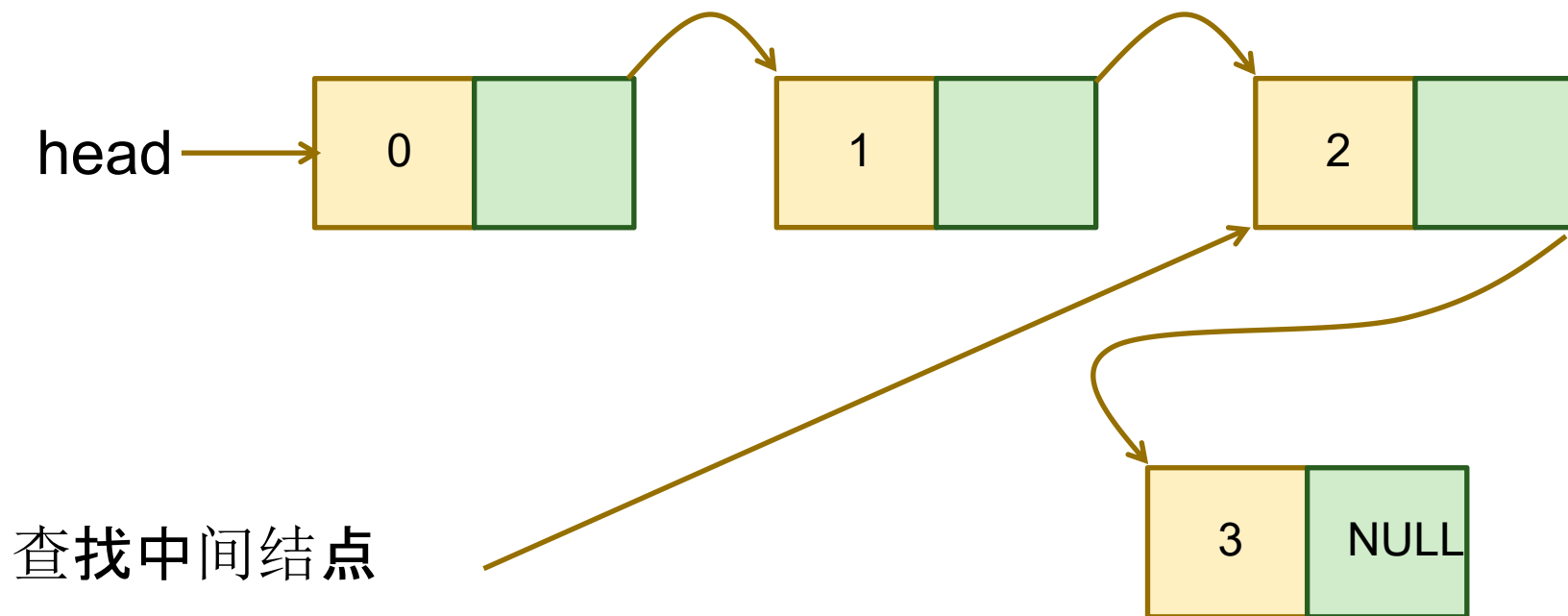
链表反转



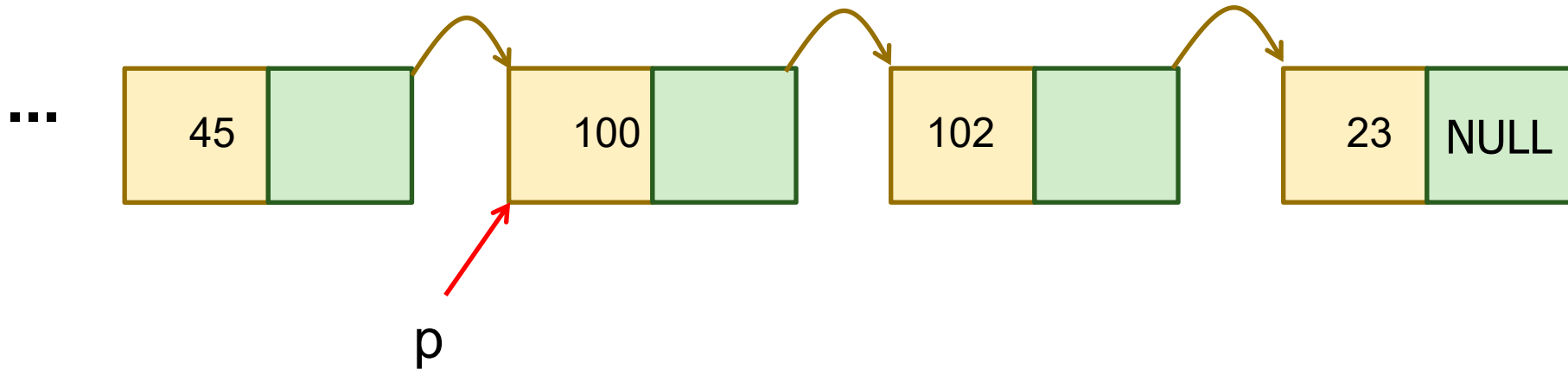
查找倒数第k个结点



查找中间结点

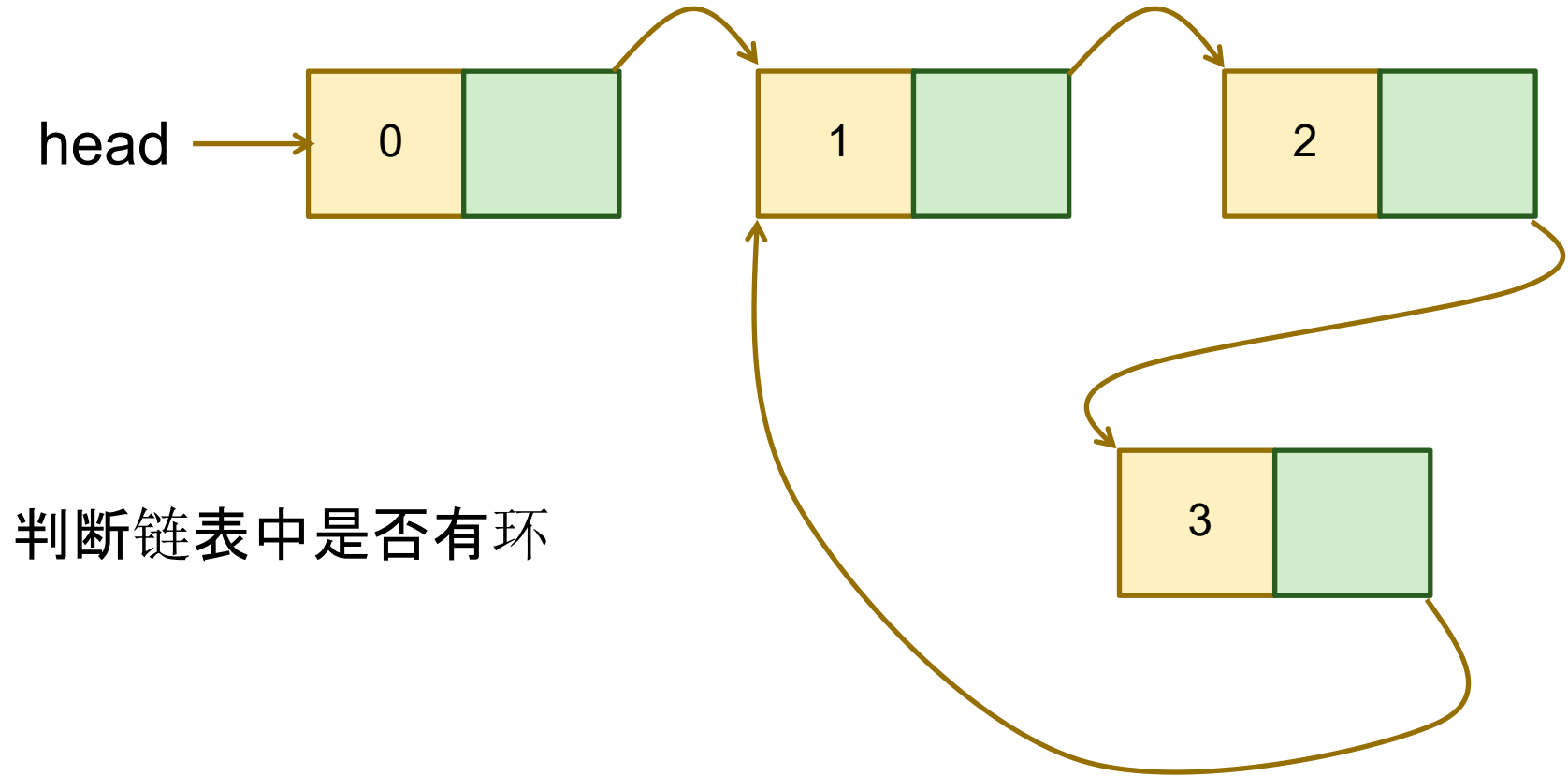


删除无头链表某个结点



不知道链表头指针，只知道中间某个节点的指针为p，如何删掉p所指向的结点？

判断是否有环



链表类 (Linked List)

```
class List {  
    private:  
        Node* head;  
        int nodeNum; //结点个数  
    public:  
        List();  
        ~List();  
        void Insert(int k, int data);  
        //在第k个位置插入新结点, 数据为data  
        void Delete(int k); //删除第k个结点  
        int Find(int k); //查找第k个结点  
        void Print(); //打印链表的所有元素  
};
```

链表类 (Linked List)

```
List::List() {  
    nodeNum = 0; //初始化结点个数  
    head = new Node(0); //创建表头  
}
```

```
List::~~List() {  
    for(int i = nodeNum; i >=1; i --) {  
        Delete(i);  
    }  
    delete head;  
}
```

链表类 (Linked List)

```
int List::Find(int k) {  
    if(k < 1 || k > nodeNum)  
        return -1; //元素不存在  
    Node *p = head;  
    for (int i=1;i<k;i++)  
        p = p->next;  
    Node *q = p->next;  
    return q->data;  
}
```

链表类 (Linked List)

```
void List::Insert(int k, int data) {  
    if(k < 1 || k > nodeNum + 1)  
        return; //插入位置不合法  
    Node *newNode = new Node(data);  
    Node *p = head;  
    for(int i=1; i<k; i++)  
        p = p->next;  
    newNode->next = p->next;  
    p->next = newNode;  
    nodeNum ++;  
}
```

链表类 (Linked List)

```
void List::Delete(int k) {  
    if(k < 1 || k > nodeNum)  
        return; //位置不合法  
    Node *p = head;  
    for (int i=1;i<k;i++)  
        p = p->next;  
    Node *q = p->next;  
    p->next = q->next;  
    delete q;  
    nodeNum --;  
}
```

链表类 (Linked List)

```
void List::Print() {  
    Node *p = head;  
    while(p) {  
        p = p->next;  
        if(p != NULL)  
            cout<<p->data<<endl;  
    }  
}
```

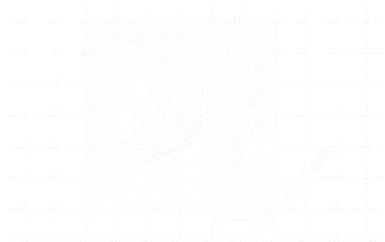
链表类 (Linked List)

```
int main() {  
    List *lt = new List();  
    lt->Insert(1, 10);  
    lt->Insert(2, 20);  
    lt->Insert(3, 30);  
    lt->Insert(4, 40);  
    lt->Print();  
  
    cout<<lt->Find(3)<<endl;  
  
    lt->Delete(3);  
    lt->Delete(1);  
    lt->Print();  
    return 0;  
}
```

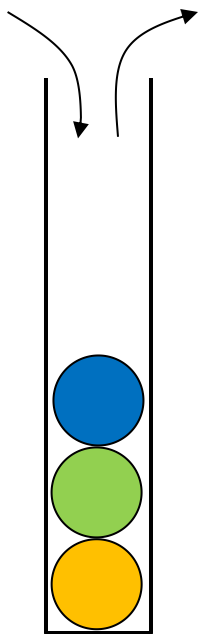
输出结果:

10
20
30
40
30
20
40

栈 (stack)



栈 (Stack)



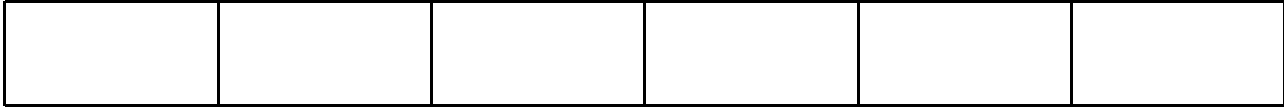
- 栈是一种线性数据结构，可以存入（插入）和取出（删除）元素
- 元素插入和删除操作都只能在栈的同一端进行，即“先入后出”
- 允许进行插入和删除操作的一端叫做**栈顶**(top)，另一端叫做**栈底**(bottom)
- 插入元素的操作称为入栈(push)，从栈中删除元素的操作称为出栈(pop)

栈

空栈

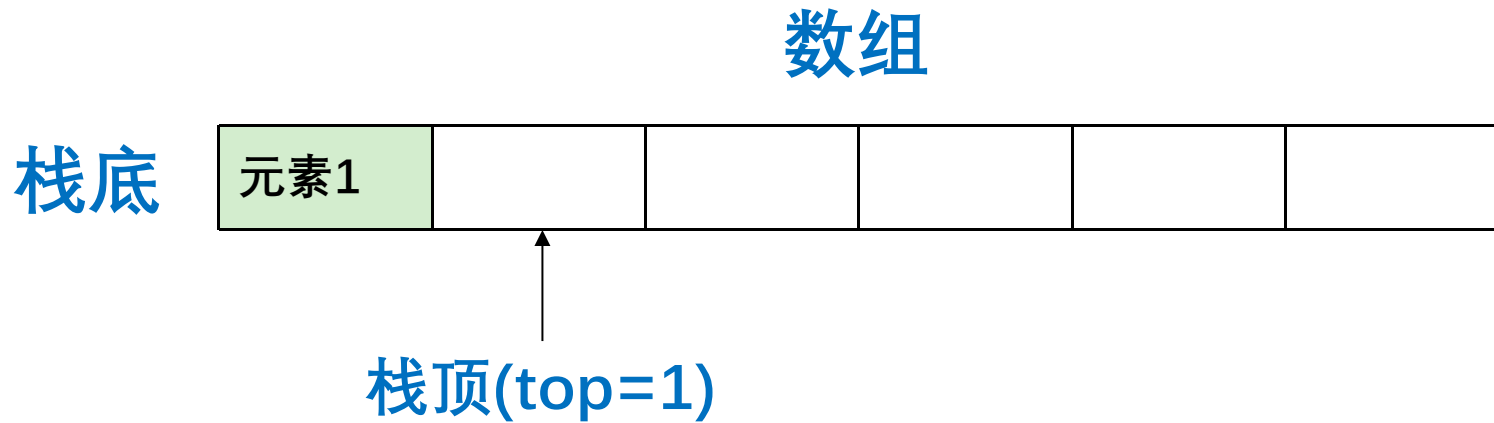
数组

栈底



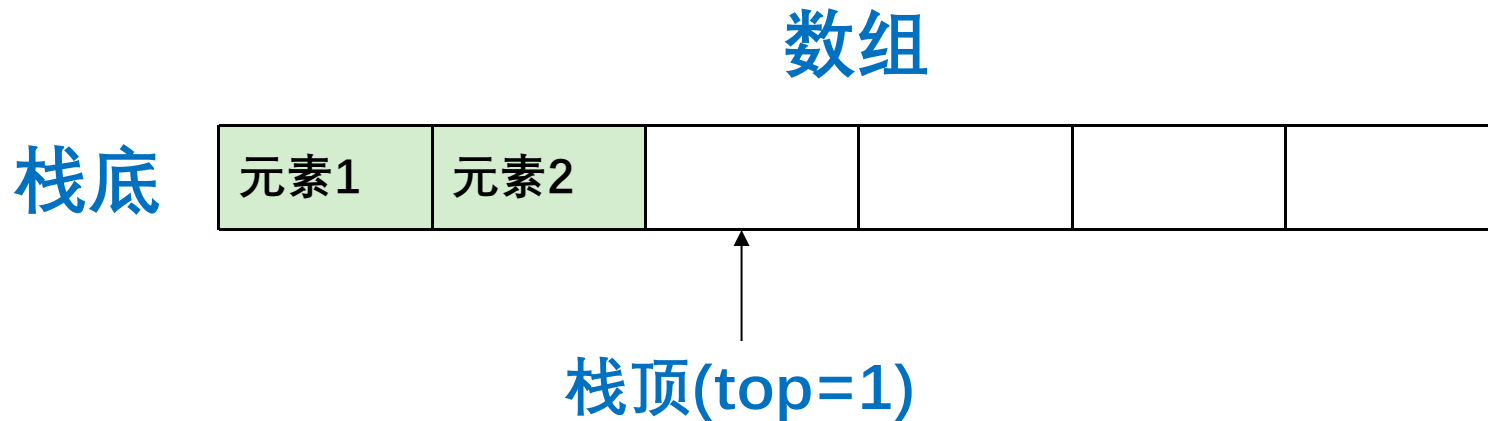
栈

插入一个元素(push)



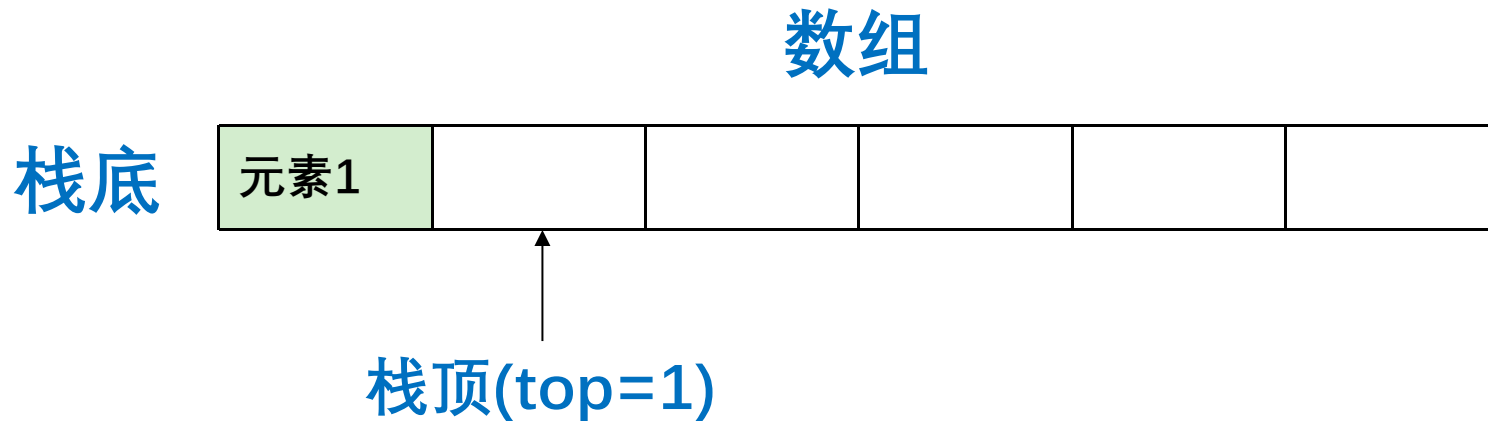
栈

再插入一个元素(push)



栈

删除一个元素(pop)



设计栈类

```
const int maxsize = 6; //栈大小，最多可存多少元素
class Stack {
    float data[maxsize]; //用来存元素的数组
    int top {0}; //栈顶
public:
    bool empty(void); //判断栈是否为空
    void push(float a); //插入一个元素
    float pop(void); //删除一个元素
};

bool Stack::empty(void) {
    return top == 0 ? true : false;
}
```

设计栈类

```
void Stack::push(float a) {
    if (top == maxsize) {
        cout<<"Stack is full!"<<endl;
        return;
    }
    data[top] = a;
    top++;
}

float Stack::pop(void) {
    if (top == 0) {
        cout<<"Stack is underflow!"<<endl;
        return 0;
    }
    top--;
    return data[top];
}
```

设计栈类

```
void main() {  
    Stack s1, s2;  
    for(int i = 1; i <= maxsize; i++)  
        s1.push(2 * i);  
    for(int i = 1; i <= maxsize; i++)  
        cout<<s1.pop()<<"  ";  
    for(int i = 1; i <= maxsize; i++)  
        s1.push(2.5*i);  
    for(int i = 1; i <= maxsize; i++)  
        s2.push(s1.pop());  
    cout<<endl;  
    do  
        cout<<s2.pop()<<"  ";  
    while (! (s2.empty()));  
}
```

设计栈类

程序运行结果：

12	10	8	6	4	2	
2.5	5	7.5	10	12.5	15	

栈的应用-中缀表达式转后缀表达式

中缀表达式: 常用的算术表达式,运算符在运算数中间,运算需要考虑优先级

$$2 + 9 / 3 - 5$$

后缀表达式: 计算机用的算术表达式,运算符在运算数后面,从左到右运算, 无需考虑优先级

$$2\ 9\ 3\ /\ +\ 5\ -$$

栈的应用-中缀表达式转后缀表达式

计算机如何计算后缀表达式？

2 9 3 / + 5 -

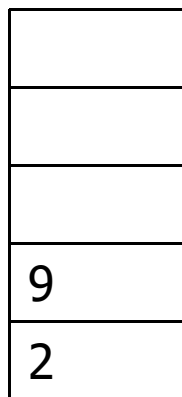
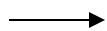
栈的应用-中缀表达式转后缀表达式

从左到右读取表达式，遇到操作数，入栈，遇到运算符就弹出相应的运算数，先弹出的作为右操作数，后弹出的作为左操作数，运算后再把结果入栈，最终结果就是栈顶的值

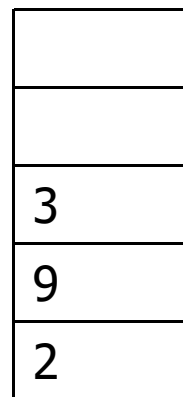
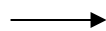
2 9 3 / + 5 -



读取2，入栈



读取9，入栈

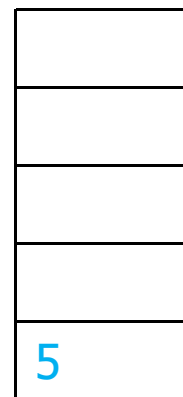
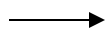
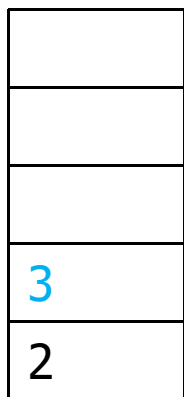


读取3，入栈

栈的应用-中缀表达式转后缀表达式

从左到右读取表达式，遇到操作数，入栈，遇到运算符就弹出相应的运算数，先弹出的作为右操作数，后弹出的作为左操作数，运算后再把结果入栈，最终结果就是栈顶的值

2 9 3 / + 5 -



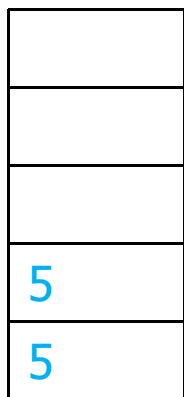
读取/, 3和9出栈,
计算9/3, 结果3入栈

读取+, 3和2出栈,
计算2+3, 结果5入栈

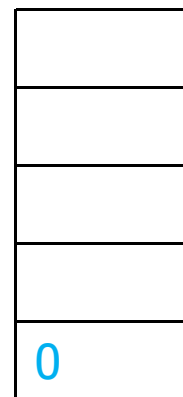
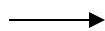
栈的应用-中缀表达式转后缀表达式

从左到右读取表达式，遇到操作数，入栈，遇到运算符就弹出相应的运算数，先弹出的作为右操作数，后弹出的作为左操作数，运算后再把结果入栈，最终结果就是栈顶的值

2 9 3 / + 5 -



读取5，入栈



读取-，5和5出栈，计算5-5，结果0入栈，为最终结果

栈的应用-中缀表达式转后缀表达式

如何将中缀表达式转换为后缀表达式？

$2 + 9 / 3 - 5 \rightarrow 2 9 3 / + 5 -$

1. 从左到右进行遍历
2. 运算数，直接输出
3. 左括号，直接压入堆栈
4. 右括号，不断弹出栈顶运算符并输出直到遇到左括号(弹出但不输出)
5. 运算符，如果栈顶是左括号，直接将运算符入栈，否则将该运算符与栈顶运算符进行比较，如果优先级高于栈顶运算符则压入堆栈，如果优先级低于等于栈顶运算符则将栈顶运算符弹出并输出，然后比较新的栈顶运算符，直到优先级大于栈顶运算符或者栈空，再将该运算符入栈
6. 如果对象处理完毕，则按顺序弹出并输出栈中所有运算符

栈的应用-中缀表达式转后缀表达式

$2 + 9 / 3 - 5 \rightarrow 2 9 3 / + 5 -$

步骤	待处理表达式	栈顶状态（低→高）	输出状态
1	$2 + 9 / 3 - 5$		
2	$+ 9 / 3 - 5$		2
3	$9 / 3 - 5$	+	2
4	$/ 3 - 5$	+	2 9
5	$3 - 5$	+ /	2 9
6	$- 5$	+ /	2 9 3
7	5	-	2 9 3 / +
8		-	2 9 3 / + 5
9			2 9 3 / + 5 -

栈的应用-中缀表达式转后缀表达式

$2*(9+6/3-5) + 4 \rightarrow 2\ 9\ 6\ 3\ /\ +\ 5\ -\ *\ 4\ +$

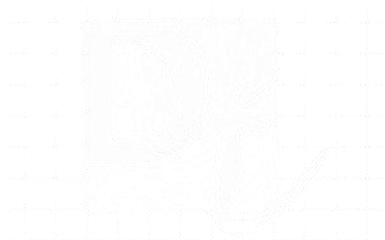
步骤	待处理表达式	栈顶状态（低→高）	输出状态
1	$2*(9+6/3-5)+4$		
2	$*(9+6/3-5)+4$		2
3	$(9+6/3-5)+4$	*	2
4	$9+6/3-5)+4$	* (	2
5	$+6/3-5)+4$	* (	2 9
6	$6/3-5)+4$	* (+	2 9
7	$/3-5)+4$	* (+	2 9 6
8	$3-5)+4$	* (+ /	2 9 6
9	$-5)+4$	* (+ /	2 9 6 3

栈的应用-中缀表达式转后缀表达式

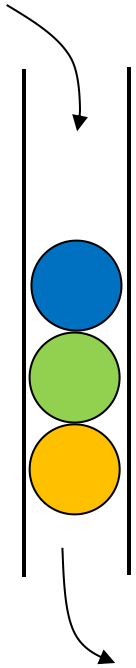
$2*(9+6/3-5) \rightarrow 2\ 9\ 6\ 3\ /\ +\ 5\ -\ *\ 4\ +$

步骤	待处理表达式	栈顶状态（低→高）	输出状态
10	5)+4	* (-	2 9 6 3 / +
11	) +4	* (-	2 9 6 3 / + 5
12	+4	*	2 9 6 3 / + 5 -
13	4	+	2 9 6 3 / + 5 - *
14		+	2 9 6 3 / + 5 - * 4
15			2 9 6 3 / + 5 - * 4 +

队列 (queue)



队列 (Queue)



- 队列是一种线性数据结构，可以存入（插入）和取出（删除）元素
- 元素插入和删除操作在不同端进行，即“先入先出”
- 允许插入的一端叫**队列尾**(rear)，允许删除的一端叫**队列头**(front)
- 插入元素的操作称为入队(queue)，从队列中删除元素的操作称为出队(dequeue)

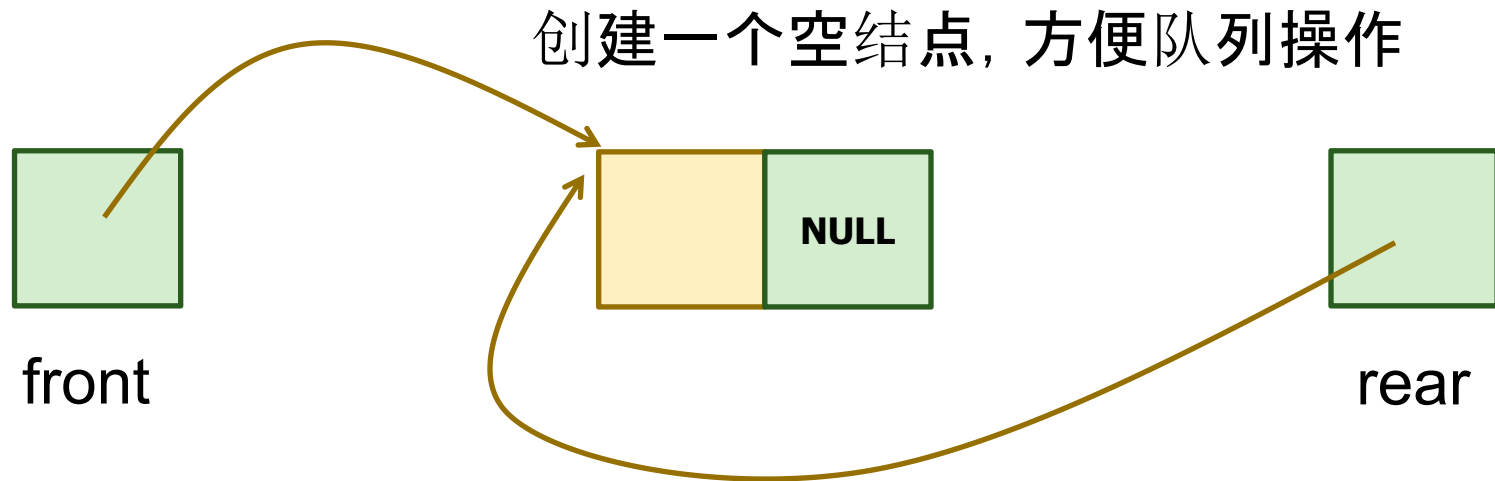
结点 (QNode)

```
class QNode{  
    public:  
        int data;  
        QNode* next;  
};
```



队列初始化

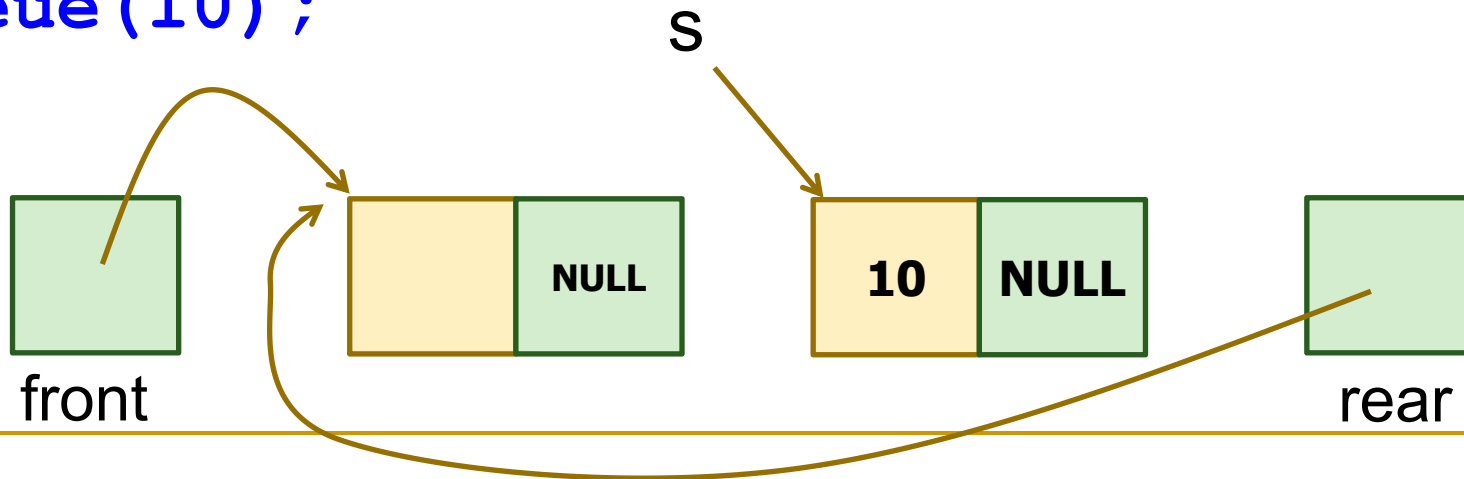
```
QNode *front; //队列头  
QNode *rear; //队列尾  
front = rear = new Qnode();  
front->next = rear;  
rear->next = nullptr;
```



入队列

```
void queue(int x) {  
    QNode *s = new QNode();  
    s->data = x;  
    s->next = nullptr;  
    rear->next = s;  
    rear = s;  
}
```

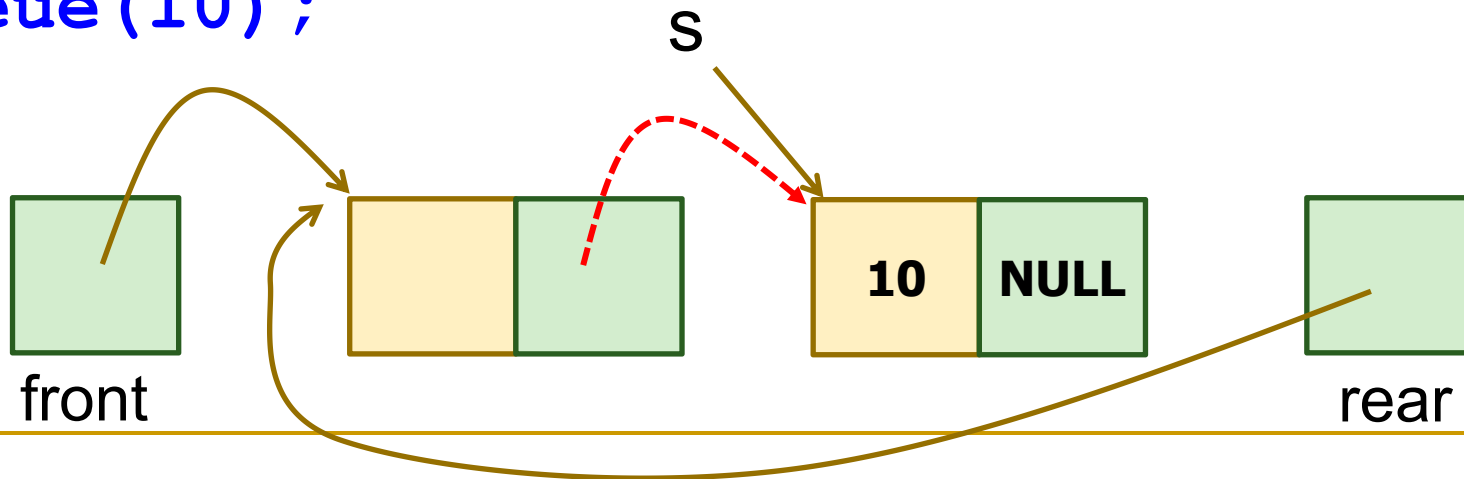
queue(10);



入队列

```
void queue(int x) {  
    QNode *s = new QNode();  
    s->data = x;  
    s->next = nullptr;  
    rear->next = s;  
    rear = s;  
}
```

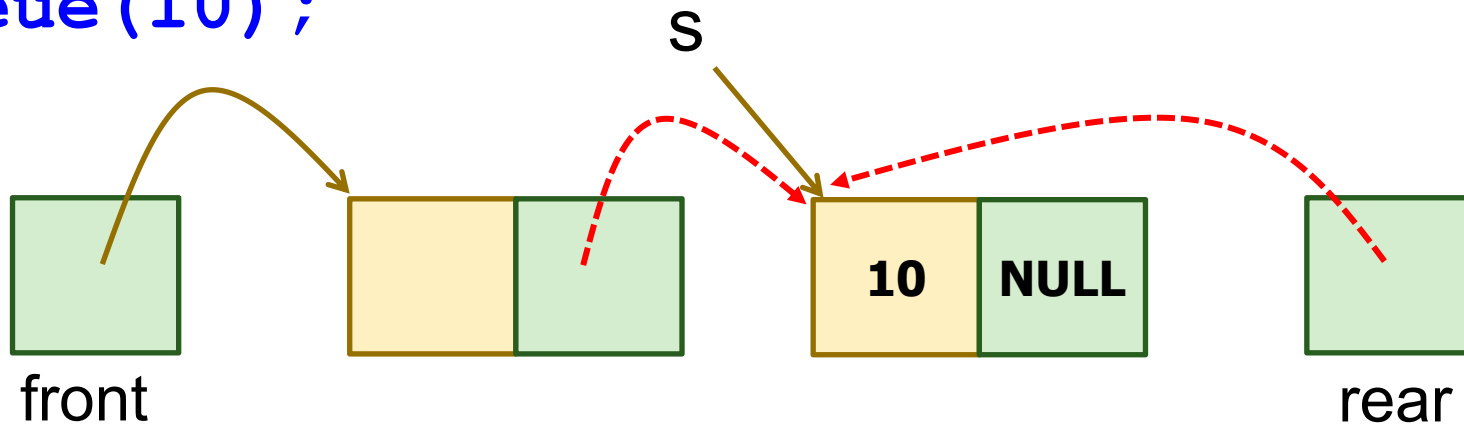
queue(10);



入队列

```
void queue(int x) {  
    QNode *s = new QNode();  
    s->data = x;  
    s->next = nullptr;  
    rear->next = s;  
    rear = s;  
}
```

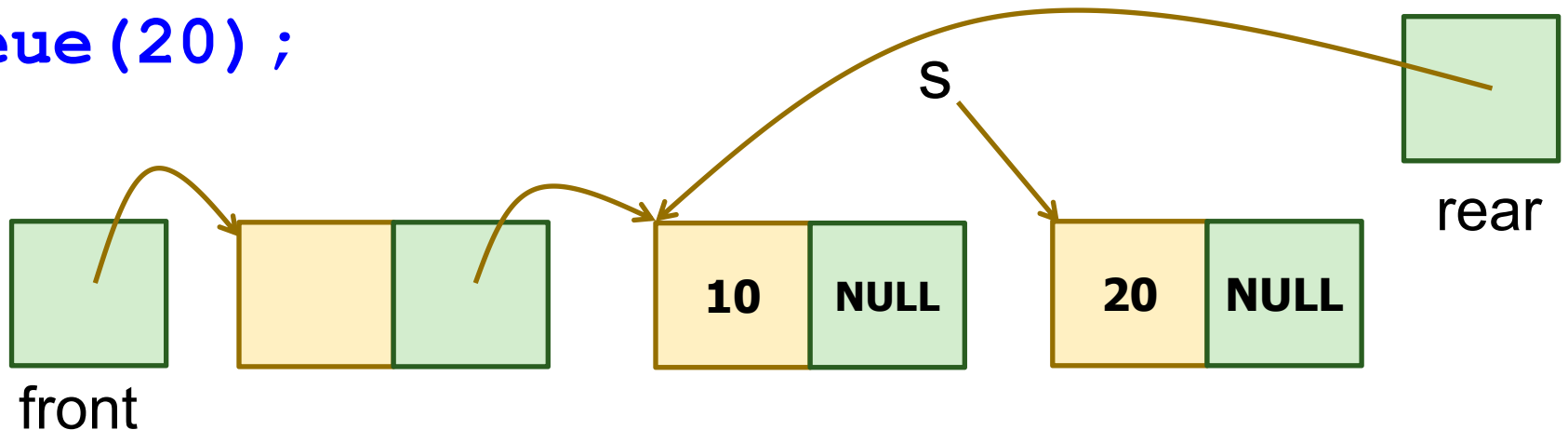
queue(10);



入队列

```
void queue(int x) {  
    QNode *s = new QNode();  
    s->data = x;  
    s->next = nullptr;  
    rear->next = s;  
    rear = s;  
}
```

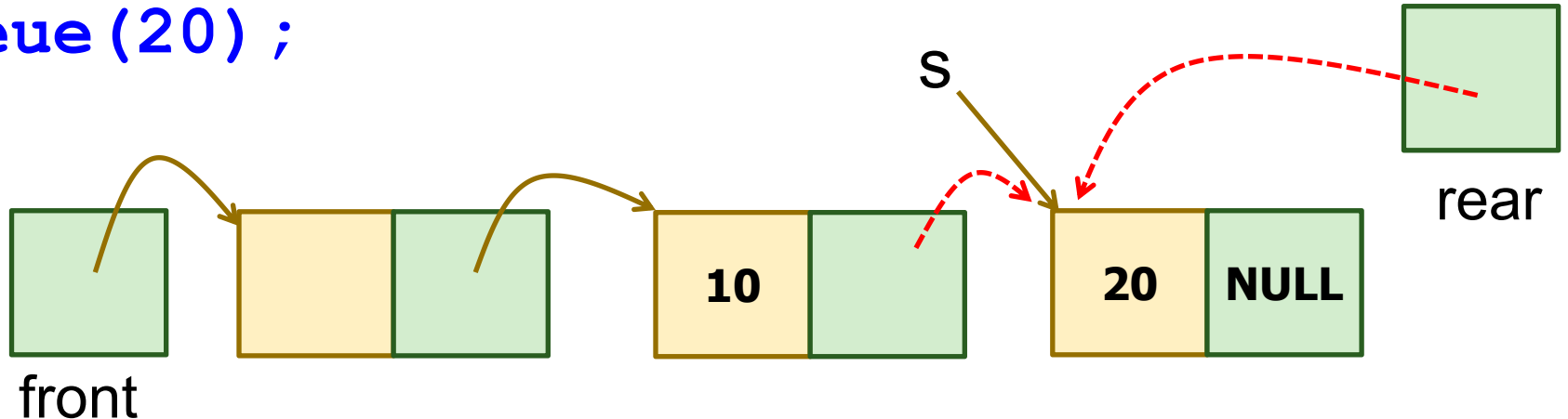
queue(20);



入队列

```
void queue(int x) {  
    QNode *s = new QNode();  
    s->data = x;  
    s->next = nullptr;  
    rear->next = s;  
    rear = s;  
}
```

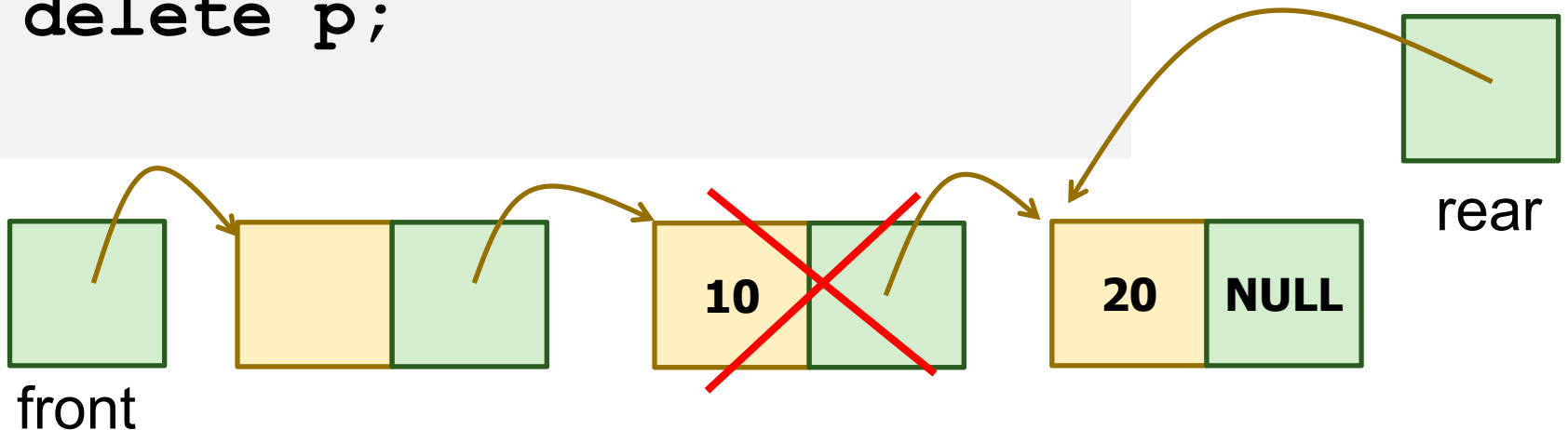
queue(20);



出队列

```
void dequeue() {  
    if (empty()) {  
        cout<<"Queue is empty";  
        return;  
    }  
    QNode *p = front;  
    front = front->next;  
    delete p;  
}
```

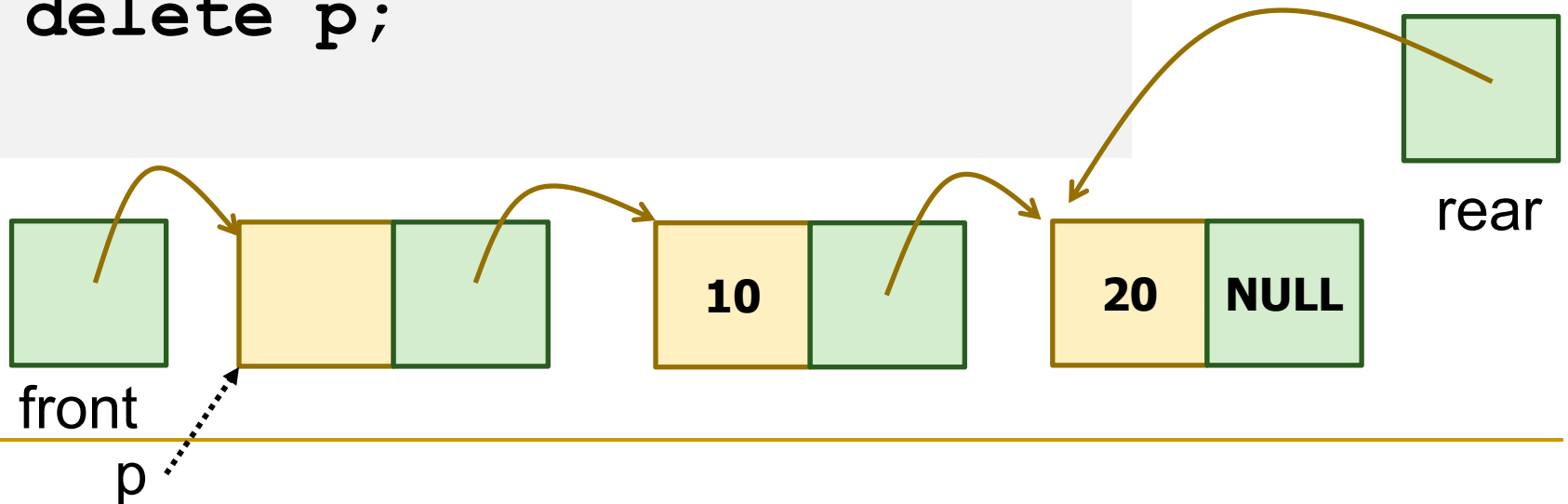
dequeue();



出队列

```
void dequeue() {  
    if (empty()) {  
        cout<<"Queue is empty";  
        return;  
    }  
    QNode *p = front;  
    front = front->next;  
    delete p;  
}
```

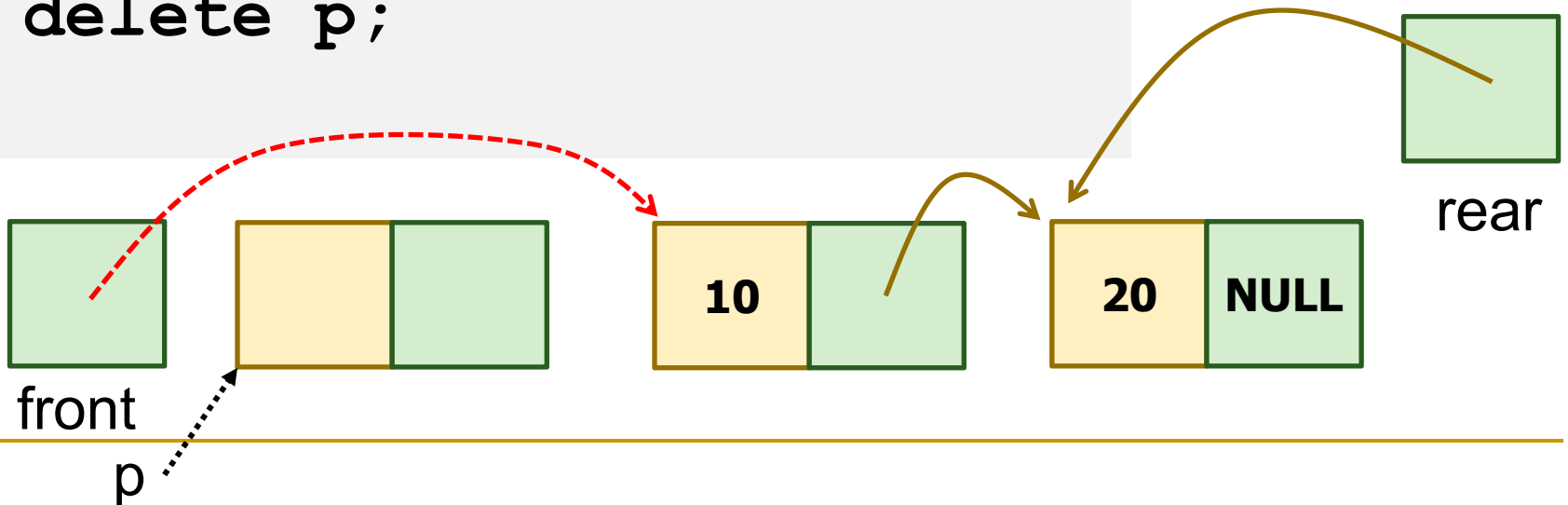
`dequeue();`



出队列

```
void dequeue() {  
    if (empty()) {  
        cout<<"Queue is empty";  
        return;  
    }  
    QNode *p = front;  
    front = front->next;  
    delete p;  
}
```

dequeue();



出队列

```
void dequeue() {  
    if (empty()) {  
        cout<<"Queue is empty";  
        return;  
    }  
    QNode *p = front;  
    front = front->next;  
    delete p;  
}
```

dequeue();



队列类实现

```
class QNode {  
public:  
    int data;  
    QNode *next;  
};
```

```
class LQueue {  
public:  
    QNode *front;  
    QNode *rear;  
    void initQueue();  
    int empty();  
    void queue(int);  
    void dequeue();  
    int getFront();  
    int getRear();  
    void printQueue();  
};
```

队列类实现

```
void LQueue::initQueue() {
    front = rear = new QNode();
    front->next = rear;
    rear->next = nullptr;
}

int LQueue::empty() { //判断队列是否为空
    return front->next == rear;
}

void LQueue::queue(int x) { //入队列
    QNode *s = new QNode();
    s->data = x;
    s->next = nullptr;
    rear->next = s;
    rear = s;
}
```

队列类实现

```
void LQueue::dequeue() {
    if (empty()) {
        cout << "LQueue is empty,can't pop";
        return;
    }
    QNode *p = front;
    front = front->next;
    delete p;
}

int LQueue::getFront() {
    return front->next->data;
}

int LQueue::getRear() {
    return rear->data;
}
```

队列类实现

```
void LQueue::printQueue() {  
    QNode *q = front->next;  
    while (q) {  
        cout << q->data << " ";  
        q = q->next;  
    }  
    cout << "\n";  
}
```

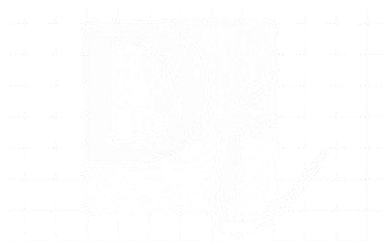
队列类实现

```
int main() {  
    LQueue L;  
    L.initQueue();  
    srand((unsigned) time(NULL));  
    for (int i = 0; i < 10; i++) {  
        L.queue(rand() % 100); //0-100之间的随机数  
    }  
    L.printQueue();  
    L.dequeue();  
    L.dequeue();  
    L.printQueue();  
    return 0;  
}
```

输出结果:

53 94 43 60 74 97 25 4 99 38
43 60 74 97 25 4 99 38

END



南方科技大学